

1 Main result & roadmap

Main Theorem (Zero false positives on the hull under *No-Flat*)

Setting. Binary 1D classification with labels $y_i \in \{\pm 1\}$ and strictly ordered inputs $x_1 < \dots < x_n$. Model: two-layer homogeneous ReLU $\Phi(\theta; \cdot)$. All requirements are (A0)–(A6) below.

Variant A (Algorithm 1: “all margin hits”). Under (A0)–(A4) and *Endpoint-tightness* (A5), running Algorithm 1 on the trained network returns only training inputs in $[x_{\min}, x_{\max}]$ (zero false positives).

Variant B (Algorithm 2: “kinks-only acceptance”). Under (A0)–(A4) and the *kinks-only acceptance rule* (A6), running Algorithm 2 returns only training inputs in $[x_{\min}, x_{\max}]$ (zero false positives).

Scope. Both variants assert no false positives on $[x_{\min}, x_{\max}]$; completeness is not claimed.

Assumptions (A0–A6)

A0 [Inherited] (Implicit-bias $\Rightarrow$ KKT). Homogeneous ReLU with logistic/exponential loss; gradient flow converges in direction to a KKT point of Euclidean max-margin.

A1 [Inherited] (White-box + KKT facts). Margin $m := \min_i |\Phi(\theta; x_i)| > 0$; full access to Φ and θ ; and the KKT conditions

$$\theta = \sum_i \lambda_i y_i \nabla_{\theta} \Phi(\theta; x_i), \quad (1.1)$$

$$y_i \Phi(\theta; x_i) \geq m, \quad \lambda_i \geq 0, \quad (1.2)$$

$$\lambda_i (y_i \Phi(\theta; x_i) - m) = 0 \quad (\text{complementary slackness}). \quad (1.3)$$

A2 [Inherited] (Univariate two-layer setup). Two-layer univariate ReLU; sorted parameter breakpoints $t_k := -b_k/w_k$. (For both variants, the attacker is given the target margin m .)

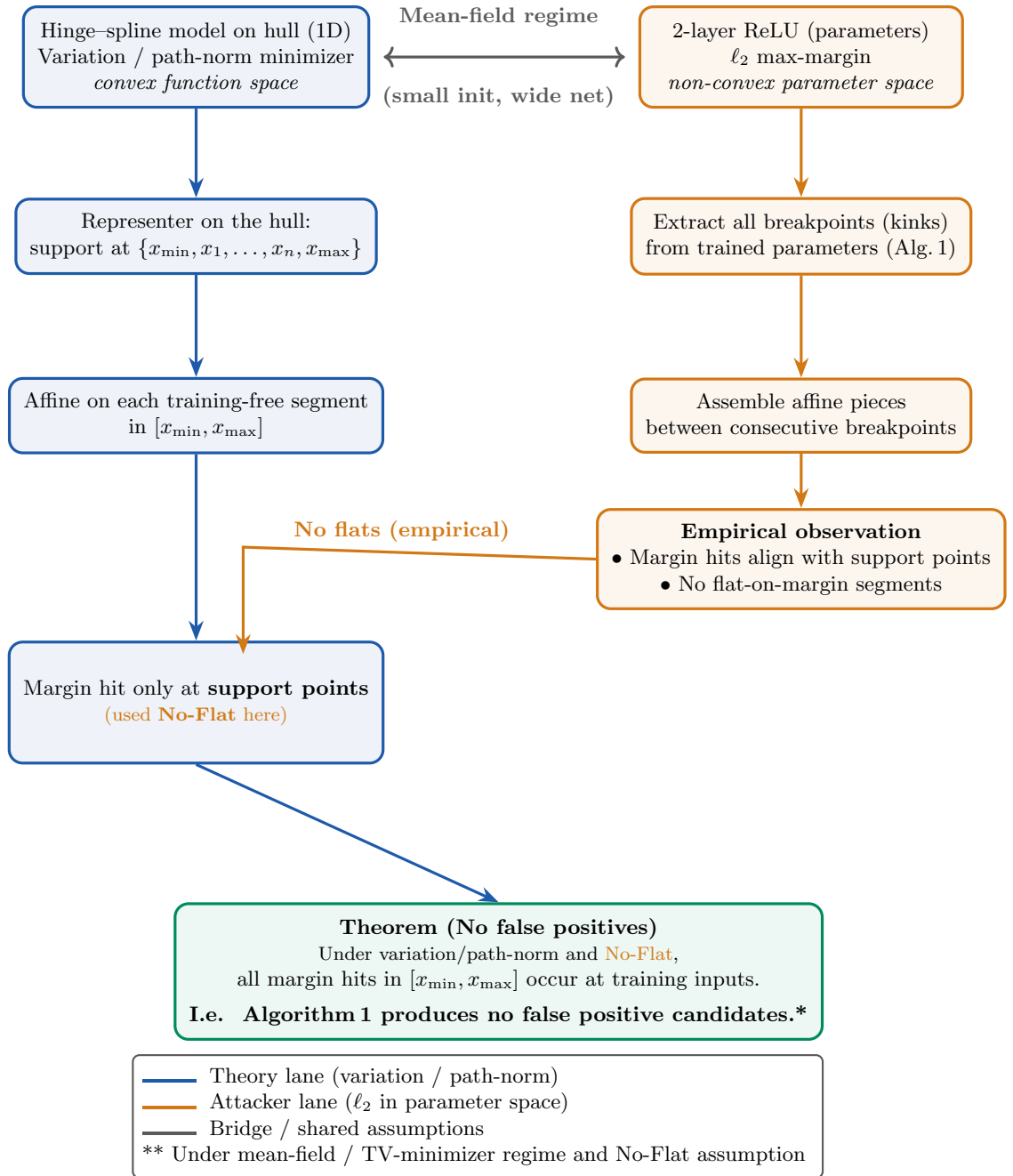
A3 [Setup] (Hull restriction). Guarantees stated on $[x_{\min}, x_{\max}] = [x_1, x_n]$.

A4 [New] (No-Flat on the margin). $\{x : |\Phi(\theta; x)| = m\}$ has no nontrivial open interval (for the chosen trained solution).

A5 [New] (Endpoint-tightness; Variant A). On any label-transition gap (x_j, x_{j+1}) with $y_j = -y_{j+1}$, both endpoints satisfy $|\Phi(\theta; x_j)| = |\Phi(\theta; x_{j+1})| = m$.

A6 [New] (Kink identification; Variant B). Accept only points with nonzero *functional* slope jump (white-box aggregation of parameter breakpoints, or black-box via one-sided differences within fixed tolerances).

A7 [Selection via convex bridge]. We analyze a function Φ that minimizes the (weighted) path/variation norm in (V) among all margin-feasible functions. This Φ serves as the analytic surrogate for the trained network under (A0)–(A2): in wide, small-initialization regimes gradient flow exhibits an implicit bias toward max-margin solutions in a convex function space (8), and for two-layer ReLU the minimal Euclidean parameter norm among realizations of a fixed Φ equals (up to normalization) its path/variation norm (6; 7). (See also the convex-duality view (5).)



2 Why standard tactics fail

Initial view and correction of the contradiction route. The target was simple to state: rule out the existence of a non-training x^* in a training-free gap with $|\Phi(x^*)| = m$. Proving the theorem therefore amounts to showing that any such interior “margin clip” is impossible. The initial plan mirrored the paper’s contradiction template: assume local optimality, then argue that an off-data margin contact permits a small, feasibility-preserving edit that reduces the norm, contradicting local optimality. A closer reading shows the appendix δ -surgery that enables such a decrease is proved only in the *flat-on-margin* regime (a whole middle segment exactly at $\pm m$ between two adjacent marginal training points, under specific sign/active-set conditions); it does *not* apply to a single non-flat interior contact. Importing it beyond that regime tacitly relies on outer-flat slope cancellations that fail in the non-flat case. Here we record why the direct parameter route stalls.

Lemma 2.1 (KKT first-order barrier). *Assume (A0)–(A1) and let θ be a KKT point with multipliers $\{\lambda_i\}$. If a direction v satisfies $D\Phi_\theta[v](x_i) = 0$ for all i , then*

$$\left. \frac{d}{dt} \right|_{t=0} \frac{1}{2} \|\theta + tv\|_2^2 = 0.$$

Proof. Using $\left. \frac{d}{dt} \right|_0 \frac{1}{2} \|\theta + tv\|_2^2 = \langle \theta, v \rangle$ and $\theta = \sum_i \lambda_i y_i \nabla_\theta \Phi(\theta; x_i)$, we obtain

$$\langle \theta, v \rangle = \sum_i \lambda_i y_i \langle \nabla_\theta \Phi(\theta; x_i), v \rangle \quad (2.1)$$

$$= \sum_i \lambda_i y_i D\Phi_\theta[v](x_i) \quad (2.2)$$

$$= 0. \quad (2.3)$$

Along the straight-line path $\theta(t) = \theta + tv$,

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \frac{1}{2} \|\theta + tv\|_2^2 = \|v\|_2^2 \geq 0, \quad (2.4)$$

so there is no negative linear term and no local decrease along any exactly-invariant surgery. $\square$

Lemma 2.2 (2-homogeneity lift-rescale cancellation). *Assume (A0)–(A1) and the standard 2-homogeneous parameterization (biases scale with weights). Let $\Delta\theta$ be any feasible lift with $\varepsilon_{\min} := \min_i \frac{y_i \Delta\Phi(x_i)}{m} \geq 0$, and choose $\alpha > 0$ with $\Phi(\alpha\theta; \cdot) = \alpha^2 \Phi(\theta; \cdot)$ and $\alpha^2 = \frac{1}{1+\varepsilon_{\min}}$. Using $\alpha^2 = (1 + \varepsilon_{\min})^{-1} = 1 - \varepsilon_{\min} + O(\varepsilon_{\min}^2)$, we expand to first order:*

Then the first-order change in $\|\cdot\|_2^2$ is zero:

$$\left. \frac{d}{d\varepsilon_{\min}} \right|_0 \|\alpha(\theta + \Delta\theta)\|_2^2 = 0.$$

Proof. Euler’s identity gives

$$\langle \nabla_{\theta} \Phi(\theta; x), \theta \rangle = 2 \Phi(\theta; x). \quad (2.5)$$

With complementary slackness, letting $S = \{i : y_i \Phi(\theta; x_i) = m\}$,

$$\langle \theta, \Delta \theta \rangle = \sum_i \lambda_i y_i \Delta \Phi(x_i) \quad (2.6)$$

$$\geq \varepsilon_{\min} m \sum_{i \in S} \lambda_i \quad (2.7)$$

$$= \frac{\varepsilon_{\min}}{2} \|\theta\|_2^2. \quad (2.8)$$

Hence, to first order in $\varepsilon_{\min}$,

$$\|\alpha(\theta + \Delta \theta)\|_2^2 = \alpha^2 (\|\theta\|_2^2 + 2\langle \theta, \Delta \theta \rangle) + O(\|\Delta \theta\|_2^2) \quad (2.9)$$

$$\approx (1 - \varepsilon_{\min}) (\|\theta\|_2^2 + \varepsilon_{\min} \|\theta\|_2^2) \quad (2.10)$$

$$= \|\theta\|_2^2 + O(\varepsilon_{\min}^2), \quad (2.11)$$

so the linear term cancels and there is no first-order gain in ℓ_2 . $\square$

On “flatten-the-middle” geometry. Emulating the paper’s δ -surgery by flattening a training-free gap fails generically in the *non-flat-on-margin* regime: sign/mask asymmetries at the gap’s tails break the cancellations that hold in the flat case. A concise counterexample is given in Appendix ??, showing the relevant cross-term is strictly positive.

Takeaway. Lemmas 2.1–2.2 block the two most direct parameter-space routes to a local ℓ_2 decrease at a KKT point (exact-invariance surgeries and lift-rescale). This justifies the pivot to the function-geometry route developed next, which delivers the soundness claims for Routes A/B without local parameter surgeries.

Mathematical preliminaries

Definition 1 (Hinge and hinge kernel). For $t \in \mathbb{R}$, the *hinge* (kernel) is

$$K(x, t) := (x - t)_+ = \max\{x - t, 0\}. \quad (2.12)$$

Proposition 1 (Hinge-integral representation on the hull). *Every continuous piecewise-linear $\Phi : [x_{\min}, x_{\max}] \rightarrow \mathbb{R}$ admits*

$$\Phi(x) = a x + c + \int_{[x_{\min}, x_{\max}]} (x - t)_+ d\mu(t), \quad (2.13)$$

for some $a, c \in \mathbb{R}$ and μ a finite signed Borel measure supported on $[x_{\min}, x_{\max}]$. Conversely, any triple (a, c, μ) with $\text{supp } \mu \subseteq [x_{\min}, x_{\max}]$ yields a continuous piecewise-linear Φ whose kinks lie in $\text{supp } \mu \cup \{x_{\min}, x_{\max}\}$.

Definition 2 (Total variation of the derivative). Let $\mathcal{K}$ be the set of kink locations of a piecewise-linear Φ . Define

$$\text{TV}(\Phi') := \sum_{u \in \mathcal{K}} |\Phi'(u^+) - \Phi'(u^-)|. \quad (2.14)$$

Proposition 2 (TV equals measure variation). *For the representation (2.13), the quantity in (2.14) equals the total-variation norm of μ :*

$$\text{TV}(\Phi') = \|\mu\|_{\text{TV}}. \quad (2.15)$$

See, e.g., (2; 4; 3).

Proposition 3 (Finite-width convex bridge: parameter $\ell_2 \Leftrightarrow$ path norm). *Consider a two-layer ReLU with biases, $\Phi_\theta(x) = \sum_{j=1}^k v_j [w_j x + b_j]_+$, with Euclidean penalties on both layers. For any fixed function Φ , minimizing the Euclidean cost over all realizations of Φ (allowing neuronwise rescalings $(v, w, b) \mapsto (\lambda v, \lambda^{-1} w, \lambda^{-1} b)$) yields the exact convex reformulation*

$$\inf_{\theta: \Phi_\theta = \Phi} \frac{1}{2} \left(\sum_j v_j^2 + \sum_j (w_j^2 + b_j^2) \right) = C_{\text{atom}} \|\Phi\|_{\text{path}}, \quad (2.16)$$

where $\|\Phi\|_{\text{path}}$ is the path norm of Φ (Def./Prop. 4), and $C_{\text{atom}} > 0$ depends only on the chosen atom normalization. See (5; 1).

Proposition 4 (1-D identification: path norm = weighted TV of Φ'). *In one dimension, writing the hinge-integral form $\Phi(x) = ax + c + \int (x - t)_+ d\mu(t)$ on the hull, the Euclidean path norm induced by (2.16) equals*

$$\|\Phi\|_{\text{path}} = \int \sqrt{1 + t^2} d|\mu|(t), \quad (2.17)$$

i.e., the weighted total variation of Φ' with weight $w(t) = \sqrt{1 + t^2}$. (If first-layer biases are penalized differently, the weight changes but remains positive and smooth.) See (6?).

Corollary 1 ((Optional) Norm equivalence by reparameterization). *Let $s = \text{asinh}(t)$. Since $dt = \sqrt{1 + t^2} ds$, one has*

$$\int \sqrt{1 + t^2} d|\mu|(t) = \int d|\tilde{\mu}|(s). \quad (2.18)$$

We keep the t -parameterization for constraints and geometry; this identity is used only to interpret $\|\Phi\|_{\text{path}}$ as an unweighted TV in a monotone coordinate. See (6?).

Proposition 5 (Existence and support restriction on the hull). *The TV-margin program in (??) attains a minimizer. Moreover, any mass at $t > x_{\text{max}}$ contributes 0 on the hull, and any mass at $t < x_{\text{min}}$ contributes an affine term absorbable into (a, c) ; hence one may restrict $\text{supp } \mu \subseteq [x_{\text{min}}, x_{\text{max}}]$ without loss. (For the weighted TV/path norm with weight $w(t) = \sqrt{1 + t^2}$, the same support restriction holds: removing off-hull mass strictly reduces the objective while leaving hull evaluations unchanged.) References: (2; 4).*

Setup (self-contained)

Definition 3 (Data, labels, hull, and gaps). Binary 1-D classification with target margin $m > 0$ and samples $(x_i, y_i) \in \mathbb{R} \times \{\pm 1\}$. After deduplication, sort

$$x_1 < x_2 < \cdots < x_n, \quad [x_{\min}, x_{\max}] = [x_1, x_n], \quad (2.19)$$

and define *gaps* (x_j, x_{j+1}) for $j = 1, \dots, n-1$ and the label-transition set $J_{\pm} := \{j : y_j \neq y_{j+1}\}$.

Proposition 6 (Equivalent model representations on the hull). *On $[x_{\min}, x_{\max}]$, the following are equivalent representations of Φ :*

$$\begin{aligned} \Phi(x) &= \sum_{k=1}^K v_k \sigma(w_k x + b_k) && \text{(parametric two-layer view)} \\ &= ax + c + \sum_{r=1}^R \alpha_r (x - t_r)_+ && \text{(finite hinge-spline view)} \\ &= ax + c + \int_{[x_{\min}, x_{\max}]} (x - t)_+ d\mu(t) && \text{(measure view on the hull).} \end{aligned} \quad (2.20)$$

Definition 4 (Problem (V): convex TV-margin on the hull). The function-space optimization we analyze is the *path-norm (weighted TV)* margin program (V). It is convex, admits linear-spline minimizers that are affine on gaps, and underlies the gap geometry used later (5; 2; 4).

$$\min_{a, c \in \mathbb{R}, \mu \in \mathcal{M}[x_{\min}, x_{\max}]} \int_{[x_{\min}, x_{\max}]} \sqrt{1 + t^2} d|\mu|(t) \quad \text{s.t.} \quad y_i \left(ax_i + c + \int (x_i - t)_+ d\mu(t) \right) \geq m \quad \forall i. \quad (V)$$

Path-norm convention. The path norm is the function-space dual of the Euclidean parameter norm for two-layer ReLU networks (6; 7). We use the weighted version from (author?) (5), which ensures that the norm is isotropic.

3 Endpoint-support representer (core engine)

Throughout we work on the hull $[x_{\min}, x_{\max}] = [x_1, x_n]$ with strictly ordered training inputs $x_1 < \cdots < x_n$, labels $y_i \in \{\pm 1\}$, gaps $I_j := (x_j, x_{j+1})$ for $j = 1, \dots, n-1$, and hinge kernel $K(x, t) = (x - t)_+$. We write the measure form of a candidate classifier as

$$\Phi(x) = ax + c + \int_{[x_{\min}, x_{\max}]} (x - t)_+ d\mu(t). \quad (3.1)$$

From the trained parameters to the function-space view. The function Φ in (3.1) is the continuous version of the two-layer ReLU network from (A2). The measure μ is the Radon–Nikodym derivative of the slope function Φ' , so that $\mu(\{t\}) = \sum_{k: -b_k/w_k=t} v_k w_k$ for a network with parameters (w_k, b_k) and breakpoints $t_k = -b_k/w_k$. We analyze the function-space version because it is convex and its minimizers are unique (up to affine drift outside the hull), whereas the parameter-space problem is nonconvex and has spurious local minima. Per (A7), we analyze the function-space minimizer Φ^* as a surrogate for the trained network $\hat{\Phi}$.

Lemma 3.1 (Two-moment identity on a gap (R1)). *Fix a training-free gap $I_j = (x_j, x_{j+1})$ and let ν be any finite signed measure supported in I_j . Define its mass and first moment on the gap by*

$$M_j := \int_{I_j} d\nu(t), \quad M_j^{(1)} := \int_{I_j} t d\nu(t). \quad (3.2)$$

Then, for every training input x_i ,

$$\int (x_i - t)_+ d\nu(t) = \begin{cases} 0, & \text{if } x_i \leq x_j, \\ x_i M_j - M_j^{(1)}, & \text{if } x_i \geq x_{j+1}. \end{cases} \quad (3.3)$$

Proof. If $x_i \leq x_j$, then $t > x_j \geq x_i$ for all $t \in I_j$, hence $(x_i - t)_+ = 0$ and

$$\int (x_i - t)_+ d\nu(t) = 0. \quad (3.4)$$

If $x_i \geq x_{j+1}$, then $(x_i - t)_+ = x_i - t$ for all $t \in I_j$, whence

$$\begin{aligned} \int (x_i - t)_+ d\nu(t) &= \int (x_i - t) d\nu(t) \\ &= x_i \int d\nu(t) - \int t d\nu(t) \\ &= x_i M_j - M_j^{(1)}, \end{aligned} \quad (3.5)$$

which proves (3.3). $\square$

Proposition 7 (Convex dual and EP certificate). *Consider the weighted path-norm program (V) on the hull with kernel $K(x, t) = (x - t)_+$ and weight $w(t) = \sqrt{1 + t^2}$ (Prop. 4). Its Lagrange dual is*

$$\max_{\lambda \in \mathbb{R}_{\geq 0}^n} m \sum_{i=1}^n \lambda_i \quad \text{s.t.} \quad \sum_{i=1}^n \lambda_i y_i = 0, \quad \sum_{i=1}^n \lambda_i y_i x_i = 0, \quad \forall t \in [x_{\min}, x_{\max}] : \left| \sum_i \lambda_i y_i (x_i - t)_+ \right| \leq w(t). \quad (3.6)$$

Let λ^* be a dual maximizer and define the dual certificate

$$g^*(t) := \sum_{i=1}^n \lambda_i^* y_i (x_i - t)_+, \quad t \in [x_{\min}, x_{\max}].$$

Then strong duality holds and any primal minimizer (a^*, c^*, μ^*) satisfies the subgradient optimality condition

$$\text{supp}(\mu^*) \subseteq \left\{ t \in [x_{\min}, x_{\max}] : |g^*(t)| = w(t) \right\}.$$

Moreover, g^* is piecewise-affine in t with breakpoints only at $\{x_1, \dots, x_n\}$; since $w(t)$ is strictly convex and smooth, the contact set $\{|g^*| = w\}$ is a finite subset of $\{x_{\min}, x_1, \dots, x_n, x_{\max}\}$.

Proof sketch. Form the Lagrangian of (V) with multipliers $\lambda_i \geq 0$ for the margin constraints and eliminate (a, c) (free variables), which enforces $\sum_i \lambda_i y_i = 0$ and $\sum_i \lambda_i y_i x_i = 0$. The measure term yields the support function of the unit ball of the weighted variation norm, giving the inequality constraint in (3.6). Strong duality follows from Slater (strict feasibility; Lemma ??). Subgradient optimality for the TV term yields support inclusion in the contact set $\{|g^*| = w\}$. Finally, $t \mapsto g^*(t)$ is piecewise-affine with kinks only at $\{x_i\}$, while w is strictly convex; hence each linear segment can meet w (or $-w$) in at most two points, which must be segment endpoints. The only possible endpoints across the hull partition are $\{x_{\min}, x_1, \dots, x_n, x_{\max}\}$. $\square$

Proposition 8 (Representer: support on data and endpoints). *There exists an optimal solution (a, c, μ^*) of (V) (with the path norm, Prop. 4) such that*

$$\text{supp}(\mu^*) \subseteq \{x_{\min}, x_1, \dots, x_n, x_{\max}\}. \quad (3.7)$$

Consequently, $\Phi^*(x) = ax + c + \int (x - t)_+ d\mu^*(t)$ is affine on every training-free gap and on the boundary gaps.

Proof. By Proposition 7, let λ^* be a dual maximizer and g^* its certificate. Any primal minimizer (a^*, c^*, μ^*) must satisfy $\text{supp}(\mu^*) \subseteq \{|g^*| = w\}$. As g^* is piecewise-affine in t with kinks only at $\{x_i\}$ and w is strictly convex, the contact set is contained in $\{x_{\min}, x_1, \dots, x_n, x_{\max}\}$. Thus there exists an optimal μ^* supported on these points, and Φ^* is affine on every training-free gap and on the two boundary gaps. $\square$

4 Gap geometry and the main theorems

We study what can happen *inside* any training-free interval (“gap”) of the input line. By the endpoint-support representer (Proposition 8) and the 1-D convex-spline geometry for program (V), any path-norm-optimal (weighted TV) classifier Φ is a linear spline and therefore *affine* on each training-free gap. We first derive two gap-wise inequalities and a tails lemma, then state a unified no-false-positive theorem with two algorithmic variants.

Assume the standing hypotheses **(A0)**–**(A4)**. For the original acceptance rule we additionally assume **(A5)** (endpoint-tightness on label transitions); for the kinks-only rule we assume **(A6)** (functional-kink identification). Let $J_{\pm} := \{j : y_j \neq y_{j+1}\}$ denote label-transition gaps.

4.1 Gap inequalities

Lemma 4.1 (Transition gap: interior strictly inside the band). ***Setting.** Let $I_j = (x_j, x_{j+1})$ be a training-free gap with opposite labels $y_j = -y_{j+1}$. Assume Φ is affine on I_j and both endpoints are marginal:*

$$y_j \Phi(x_j) = m, \quad (4.1)$$

$$y_{j+1} \Phi(x_{j+1}) = m. \quad (4.2)$$

Conclusion. For every $x \in (x_j, x_{j+1})$,

$$|\Phi(x)| < m. \quad (4.3)$$

Proof. Write $x = (1-t)x_j + tx_{j+1}$, $t \in (0, 1)$. By affinity,

$$\Phi(x) = (1-t)\Phi(x_j) + t\Phi(x_{j+1}) \quad (4.4)$$

$$= (1-t)y_j m + t(-y_j)m \quad (4.5)$$

$$= (1-2t)y_j m. \quad (4.6)$$

Hence $|\Phi(x)| = |1-2t|m < m$. $\square$

Lemma 4.2 (Same-label gap: interior strictly beyond the band). ***Setting.** Let $I_j = (x_j, x_{j+1})$ be a training-free gap with identical labels $y_j = y_{j+1} = y \in \{\pm 1\}$. Assume Φ is affine on I_j and*

$$y \Phi(x_j) \geq m, \quad (4.7)$$

$$y \Phi(x_{j+1}) \geq m, \quad (4.8)$$

together with No-Flat-on-Margin (A4). Conclusion. For all $x \in (x_j, x_{j+1})$,

$$y \Phi(x) > m. \quad (4.9)$$

Proof. Let $x = (1-t)x_j + tx_{j+1}$, $t \in (0, 1)$. Then

$$y \Phi(x) = (1-t)y \Phi(x_j) + ty \Phi(x_{j+1}) \quad (4.10)$$

$$\geq m. \quad (4.11)$$

If equality held for some interior x , we would have

$$y \Phi(x_j) = m, \quad (4.12)$$

$$y \Phi(x_{j+1}) = m, \quad (4.13)$$

so the segment would lie flat at level m on the gap, contradicting (A4). $\square$

Lemma 4.3 (Boundary gaps and tails). *On each half-line $(-\infty, x_1)$ and (x_n, ∞) , any TV-optimal solution of (V) is affine. Consequently,*

(i) *there are no functional kinks outside the hull,* (4.14)

(ii) *an affine tail intersects $\{\pm m\}$ at most once per sign (hence at most two per tail).* (4.15)

Proof. By Prop. 8, $\text{supp}(\mu) \subseteq \{x_{\min}, x_1, \dots, x_n, x_{\max}\}$; thus the tails are affine and have no kinks. If a tail is $ax + b$, then

$$ax + b = m, \quad (4.16)$$

$$ax + b = -m, \quad (4.17)$$

each has at most one solution. $\square$

4.2 Unified main theorem (two acceptance rules)

Theorem 4.4 (Zero false positives on the hull under No-Flat). ***Setting.** Binary 1-D classification with strictly ordered inputs $x_1 < \dots < x_n$, labels $y_i \in \{\pm 1\}$, and model $\Phi(\theta; \cdot)$ (two-layer homogeneous ReLU). Assume **(A0)–(A4)** throughout.*

Variant A (Algorithm 1: “all margin hits”). *Under **(A0)–(A4)** and endpoint-tightness **(A5)** on all $j \in J_{\pm}$, Algorithm 1 returns only training inputs in $[x_{\min}, x_{\max}]$ (zero false positives on the hull).*

Note. *Variant A is a hull-only guarantee; off-hull affine tails can intersect $\{\pm m\}$ without kinks (Lemma 4.3).*

Variant B (Algorithm 2: “kinks-only acceptance”). *Under **(A0)–(A4)** and **(A6)**, Algorithm 2 returns only training inputs in $[x_{\min}, x_{\max}]$ (zero false positives on the hull).*

Proof (both variants). Per (A7), we analyze the function-space minimizer Φ^* of the path-norm program (V). By Proposition 8, take a path-norm-optimal Φ that is affine on each training-free gap.

Variant A. On same-label gaps, Lemma 4.2 gives $y\Phi(x) > m$ in the interior. On transition gaps with both endpoints tight, Lemma 4.1 gives $|\Phi(x)| < m$ in the interior. Thus any hull point with $|\Phi(x)| = m$ is a training input (or an endpoint).

Variant B. Algorithm 2 accepts only *functional kinks* that lie on the margin. Interior points of any gap are either non-marginal (by Lemmas 4.2–4.1) or marginal but not kinks (affinity), hence excluded. Therefore, accepted hull points are training inputs (or endpoints) in both variants. $\square$

Remarks.

- **Endpoint tightness for Variant A.** Lemma 4.1 leverages tightness at both ends of a transition gap; this can be assumed **(A5)**, derived via a mild dual non-degeneracy / vanishing strictly-convex tie-break, or enforced by selection.
- **Outside the hull.** By Lemma 4.3, tails are affine and can cross $\{\pm m\}$ at most twice per tail. Variant B remains globally zero-FP (only kinks are accepted); Variant A may require an outward-tail selection to avoid off-hull crossings.

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- **Approximate version.** If Φ^* is TV-optimal and

$$\|\Phi - \Phi^*\|_\infty \leq \delta, \quad (4.18)$$

$$\Delta > \delta, \quad (4.19)$$

where Δ is the smallest gap slack implied by Lemmas 4.1–4.2, then conclusions hold up to tolerance (replace equalities by ε -balls in value/position with $\varepsilon \asymp \delta$).

5 Attack-facing guarantees (Algorithm 2): scanning breakpoints yields no false positives

This section formalizes what an *attacker* can reliably infer by scanning the *parameter breakpoints* $t_k := -b_k/w_k$ of a trained two-layer ReLU network and testing locally whether those locations behave like *on-margin kinks* of the learned function.

Attacker’s knowledge of the margin. Throughout this section we assume the attacker is given the target margin level m (Assumption A2).

All statements *build on* the main results of Chapter 4: endpoint-support representer (Prop. 8), affine-on-gaps geometry, and the gap inequalities (Lemmas 4.1–4.2), as well as the tail behavior (Lemma 4.3).

5.1 Attacker model and candidate filter

Definition 5 (Attacker capabilities and candidate set). The attacker does not know the training inputs nor the hull $[x_{\min}, x_{\max}]$. They can: (i) enumerate all *parameter breakpoints* $t_k = -b_k/w_k$ from the trained parameters, (ii) query $\Phi(x)$ at any x , and (iii) approximate one-sided slopes $\Phi'(t^\pm)$ (e.g., one-sided finite differences). Given tolerances $\varepsilon, \tau > 0$, define the candidate set

$$\mathcal{C}_{\varepsilon, \tau} := \left\{ t : \left| |\Phi(t)| - m \right| \leq \varepsilon \text{ and } \left| \Phi'(t^+) - \Phi'(t^-) \right| > \tau \right\}. \quad (5.1)$$

We will show that, under our function-space model and **(A4)** (No-Flat-on-Margin), every $t \in \mathcal{C}_{\varepsilon, \tau}$ coincides (up to the tolerance) with a training input; i.e., the scan yields *no false positives*.

5.2 What is locally testable at a breakpoint

Lemma 5.1 (Derivative jump equals aggregate hinge amplitude). *Let $t \in \mathbb{R}$. For a two-layer ReLU*

$$\Phi(x) = \sum_{k=1}^K v_k [w_k x + b_k]_+, \quad (5.2)$$

the net slope jump satisfies

$$\Phi'(t^+) - \Phi'(t^-) = \sum_{k: -b_k/w_k=t} v_k w_k, \quad (5.3)$$

with the convention that neurons with $w_k = 0$ contribute 0.

Definition 6 (Active vs. inactive parameter breakpoint). A location t is an *active breakpoint* of Φ if

$$\Delta'(t) := \Phi'(t^+) - \Phi'(t^-) \neq 0; \quad (5.4)$$

otherwise it is *inactive*.

Operational note. In practice one estimates $\Delta'(t)$ numerically via one-sided finite differences and thresholds by τ to remove breakpoints whose individual neuron kinks *cancel* in aggregate.

5.3 Structure recalled from Chapter 4

By the endpoint-support representer (Prop. 8) we may assume—without loss of optimality—that all functional kinks occur *only* at training inputs or the hull endpoints. Consequently, Φ is *affine* on each training-free gap (x_j, x_{j+1}) , so there are *no active breakpoints inside gaps*; and Φ has *affine tails* outside the hull, hence no off-hull kinks (Lemma 4.3). We also use the gap inequalities: on label-transition gaps with marginal endpoints the interior lies strictly inside $(-m, m)$ (Lemma 4.1), and on same-label gaps the interior lies strictly beyond $\pm m$ under (A4) (Lemma 4.2).

5.4 Soundness of Algorithm 2 (exact case)

Theorem 5.2 (Algorithm 2: no false positives (global)). *Assume (A0)–(A4) and (A6). Let Φ be a TV-optimal solution in endpoint-support form. If t is active and*

$$|\Phi(t)| = m, \quad (5.5)$$

then t coincides with a training input. In particular, Algorithm 2 (kinks-only acceptance) produces no false positives on the hull and—since there are no kinks off-hull—globally.

Proof. Per (A7), we analyze the function-space minimizer Φ^* of the path-norm program (V). Inside any gap, Φ is affine, hence

$$\Phi'(t^+) - \Phi'(t^-) = 0, \quad (5.6)$$

so an *active* t cannot lie in a gap. Moreover, the gap inequalities exclude interior on-margin hits. Off-hull, Lemma 4.3 gives no kinks. Thus any active on-margin point must be a support point—i.e., a training input (endpoints coincide with data support after deduplication). $\square$

5.5 Quantitative robustness (tolerances and approximation)

Lemma 5.3 (Positive slack inside gaps). *Let Φ^* be a TV-optimal solution satisfying (A4). Define the gap slack*

$$\Delta := \min \left\{ \min_{j: y_j = y_{j+1}} \min_{x \in (x_j, x_{j+1})} (y_j \Phi^*(x) - m), \right. \\ \left. \min_{j: y_j = -y_{j+1}} \min_{x \in (x_j, x_{j+1})} (m - |\Phi^*(x)|) \right\}. \quad (5.7)$$

Then $\Delta > 0$.

Theorem 5.4 (Algorithm 2: sound under tolerances and approximation). *Let Φ^* and Δ be as above and suppose*

$$\|\Phi - \Phi^*\|_\infty \leq \delta. \quad (5.8)$$

Fix tolerances with $0 < \varepsilon < \Delta - \delta$ and any $\tau > 0$. Then every $t \in \mathcal{C}_{\varepsilon, \tau}$ (Def. 5) coincides (up to the value tolerance) with a training input. Hence Algorithm 2 remains sound (no false positives).

Proof. For any interior x in a gap, the exact minimizer satisfies

$$||\Phi^*(x)| - m| \geq \Delta. \quad (5.9)$$

Uniform δ -closeness and $\varepsilon < \Delta - \delta$ exclude passing the on-margin test inside gaps; and the kink test fails there since Φ is affine. Off-hull there are no kinks. Thus any retained t must be at a data support. $\square$

Takeaway. Under (A0)–(A4) and (A6), scanning parameter breakpoints and keeping only *active on-margin* ones (Algorithm 2) yields *no false positives* on the hull—and, since there are no off-hull kinks, *globally*.

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